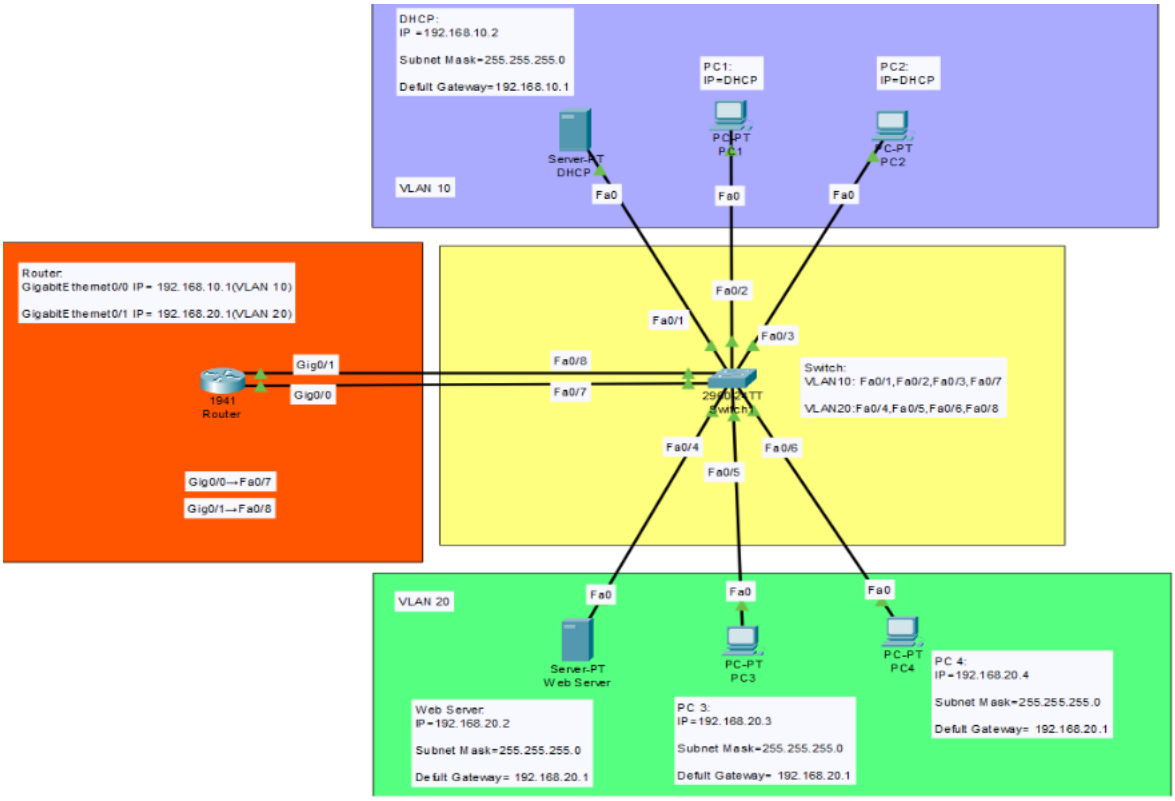
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Introduction

For this assignment it is required to design and implement a small network using VLANs in Cisco Packet Tracer For this report I will explain, describe, and justify the network setup. explanation of how the kernel manages processes and how the CPU executes the instructions which also testing results.

Network Topology Diagram



Switch Ports	Device &Port	VLAN	IP addressing
Fa0/1	To DHCP server (Fa0)	VLAN 10	IP =192.168.10.2 Default Gateway=192.168.10.1
Fa0/2	PC 1(Fa0)	VLAN 10	Receives IP from DHCP server
Fa0/3	PC2 (Fa0)	VLAN 10	Receives IP from DHCP server
Fa0/4	Web server (Fa0)	VLAN 20	IP=192.168.20.2 Default Gateway=192.168.20.1
Fa0/5	PC3(Fa0)	VLAN 20	IP=192.168.20.3 Default Gateway=192.168.20.1
Fa0/6	PC4(Fa0)	VLAN 20	IP=192.168.20.4 Default Gateway=192.168.20.1
Fa0/7	Router (Gig0/0)	VLAN 10	GigabitEthernet0/0 IP= 192.168.10.1

Fa0/8	Router (Gig0/1)	VLAN 20	GigabitEthernet0/1 IP= 192.168.20.1

Network Design, Justification of addressing schemes and VLAN configuration

For this network I have used one switch (2960), one router (1941), one DHCP, one Web server and 4 Pcs. this network is consisting of two Vlan's 10 and 20. VLAN10 contains the DHCP server that serves, and its purpose is to assign Ip addresses automatically to PC 1 and PC2 rather than a static which minimises the possibility of human error. at The DHCP has a pool with a starting Ip address of 192.168.10.3 and a maximum number of users of 10. This is enough to cover PC 1 and PC 2 and room for extension if that would be needed. While on the other hand, VLAN 20 hosts the Web server (192.168.20.2) and PC 3 and PC 4 were each of the PCs in Vlan 20 has their own static IP address (192.168.20.3 and 192.168.20.4) I chose the Ip addresses numbered in order which helps in making it structured and functional, all these devices were connected using a straight though copper cable. (See Image and table upstairs). When configuring the router to and dedicating a port to each Vlan these Ip addresses where entered (192.168.10.1) for vlan 10 and (192.168.20.1) The purpose of choosing these Ip addresses is that the third octet matches with VLAN 10 and 20 making it easier to remember and identify in case of issues. **[Amin Amini, Lecture 10]**

This Network uses star topology with the legacy inter-vlan routing configuration that uses a layer 2 network outline, as it consistently uses 2 separated VLANs, the advantages of this is that it will improve the security as relaying one individual Vlan to send sensitive data can have a risk of potentially leaking the information to unauthorised people to see it. Another strength to the use of 2 VLANs is that it helps reducing the broadcast traffic between the devices making it smoother to operate. This method will also allow for improvement as I can add more devices and function easily due to it being designed to be simple and easy to operate. [**Cisco Networking Academy, 2020]**

Operating System Kernel

The operating system kernel is the core part that is responsible for managing the system resources it allows communication between applications and the hardware such as memory. Its main duties and responsibilities are to process management, device controlling, managing system call and handling memory.

Applications cannot directly access or interact with hardware therefore it uses system calls instead which act as a controlled way into the kernel and allows for communication/access with network interface card, storage devices, and memory. This

happens by having the kernel translate the system calls request into hardware instructions to ensure secure access which enables functions without needing to directly communicate with the hardware
[JumpCloud, 2025] [Amin Amini, Lecture 4]

Including an example of my own work related to kernel is that when a service such as the Web server or DHCP server run on a system the kernel will then manage it as a process, and when the service start it will issue a system call. The kernel creates a PCB (process control Block) for each process that have been created that contains information such as priority, memory address, open files, and process ID. The kernel will then go to scheduling which process the CPU will execute first and this will ensure that the web server processes receive enough time to respond to the request of the network. **[Amin Amini, Lecture 3]**

CPU Execution

The CPU executes all programs including the DHCP and Web server using the fetch-decode-execute cycle. This cycle keeps running billion times per second, regardless of whether the network requests or not.

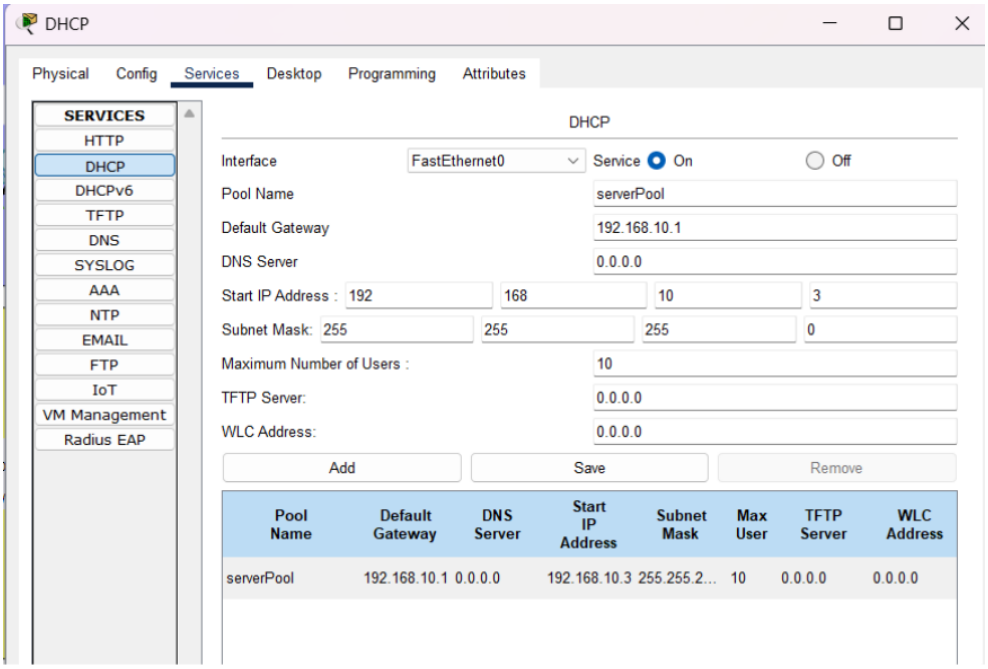
1. Fetch: The CPU executes the instruction using the Fetch-Decode- Execute Cycle. The CPU fetches he instruction from the memory that the DHCP or Web server program when requesting an Ip address or when trying to access a web page.
2. Decode: The Control unit works out what the instructions means to understand what action is required. So, for the DHCP it would be decode the request from PC1 or 2 for an Ip address or a request to read the file.
3. Execute: the CPU perform the operation whether, in example in my case it is assigning the Ip address requested from the DHCP or loading the webpage from the Web server **[James WoodHouse,2025]**

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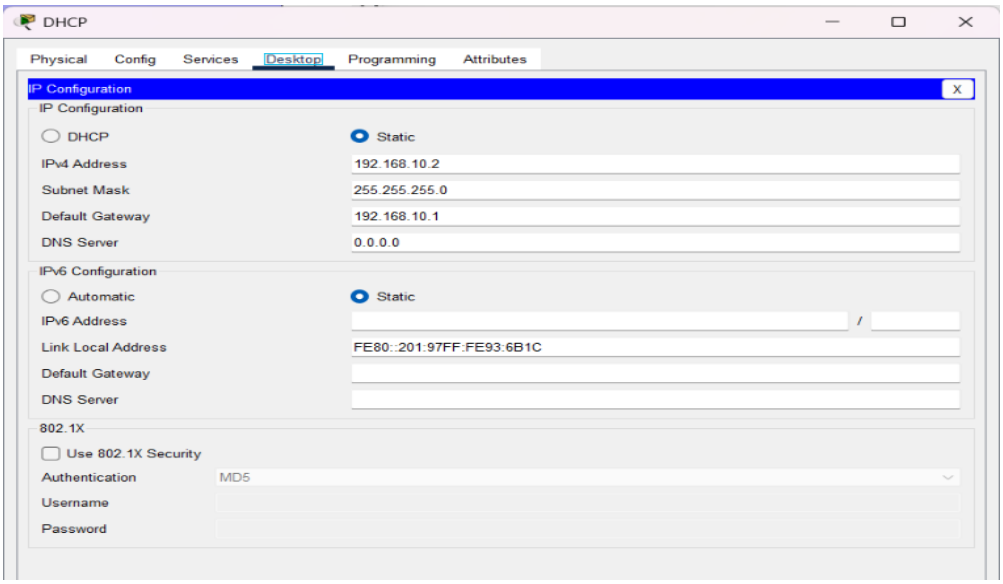
Configuration details of the DHCP server and Web server

DHCP

DHCP server pool & configuration

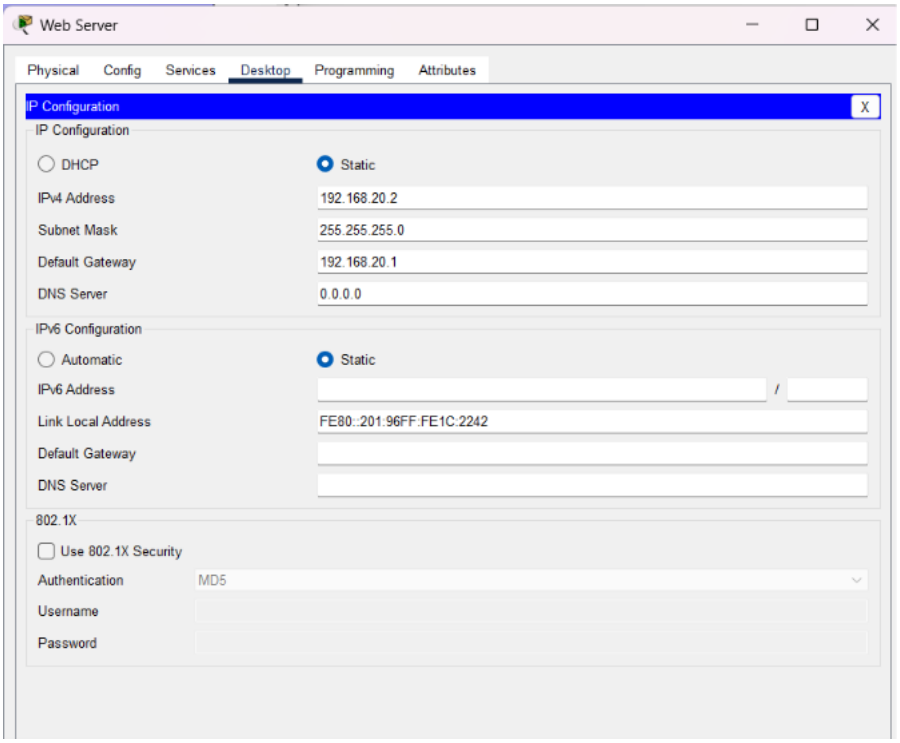


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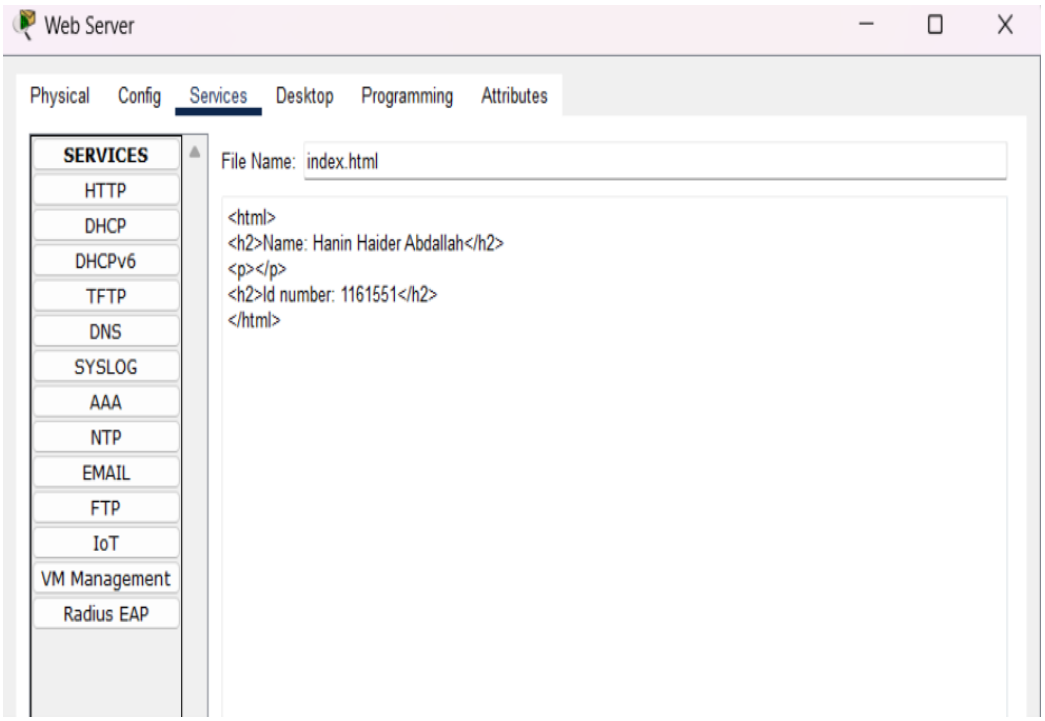


Web server

Configuration &website



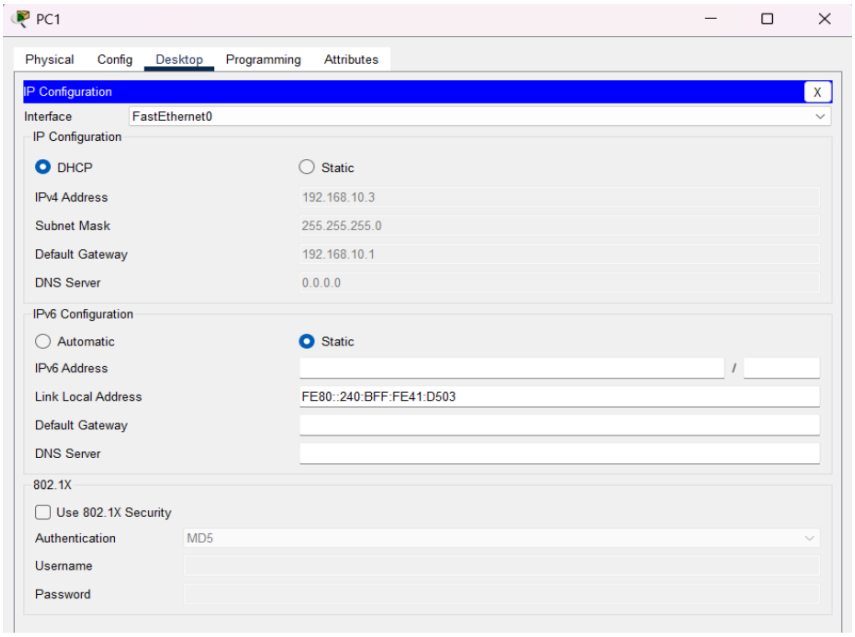
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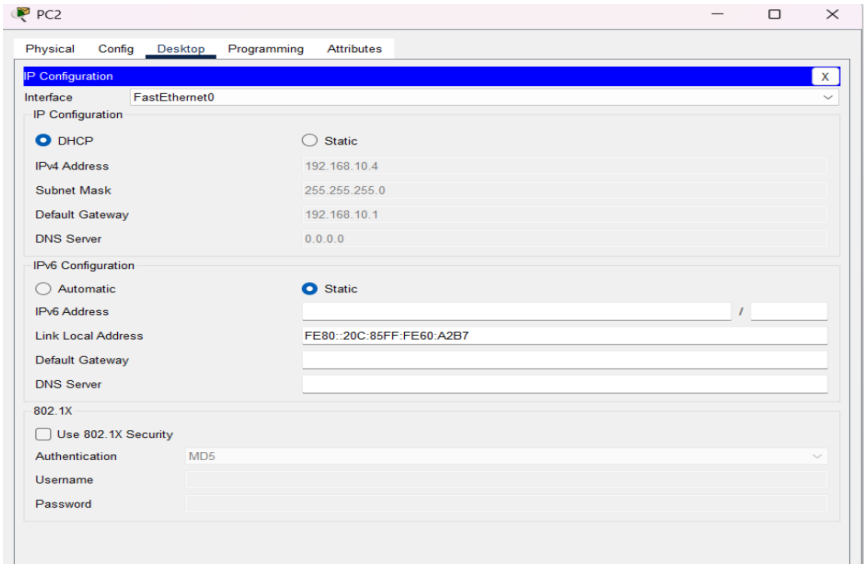
Testing Result

PCs IP addresses

PC1&2 receives automatically from DHCP

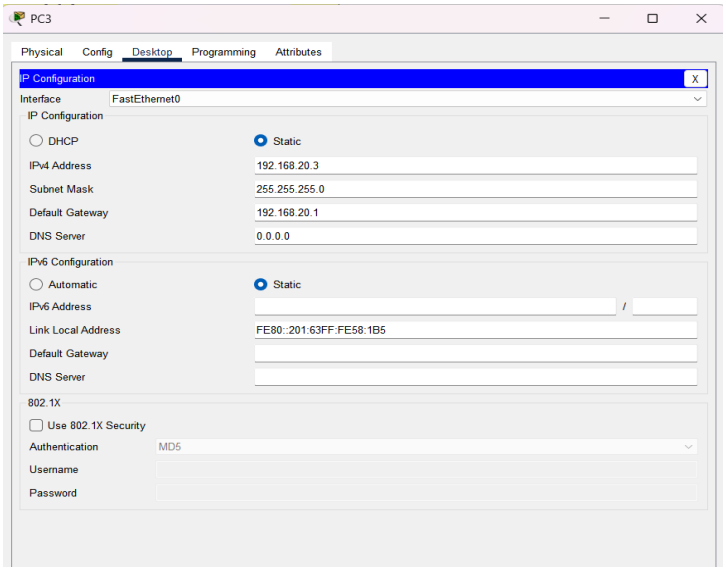


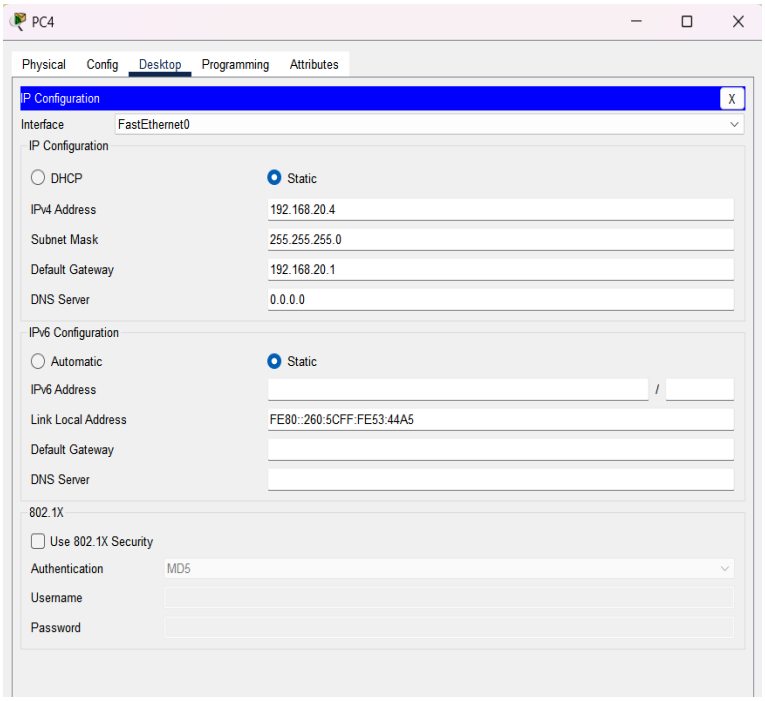
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PC3&4 static IP address

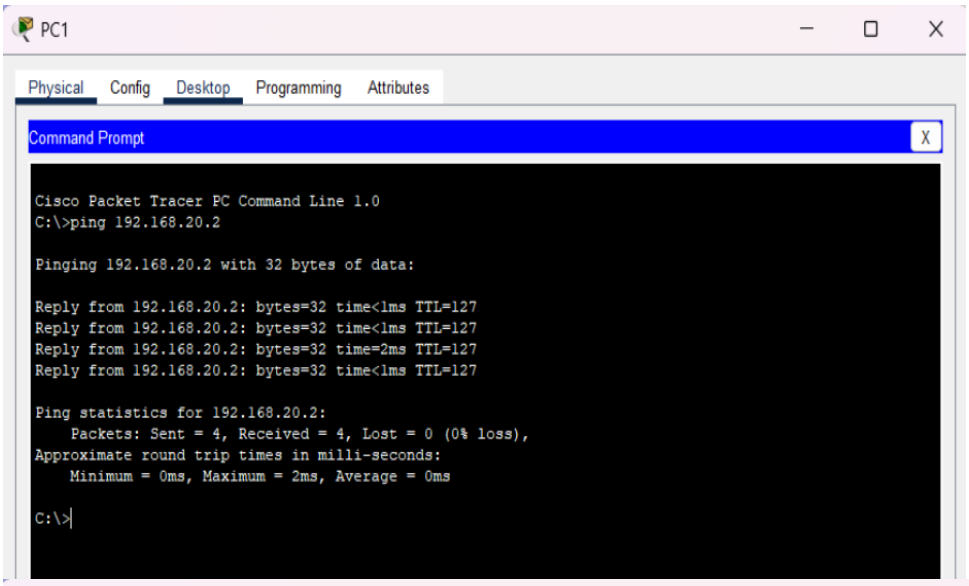
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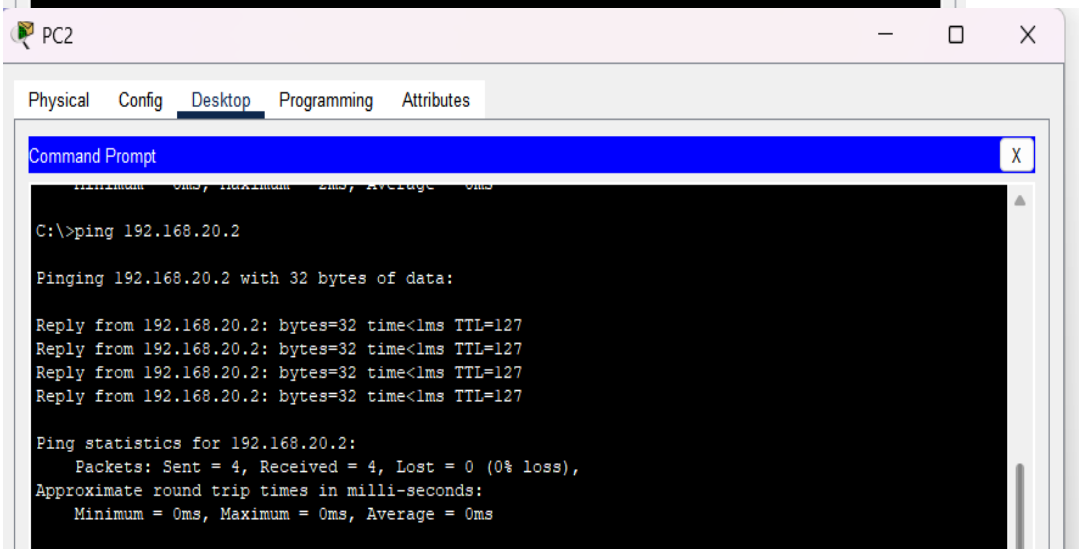


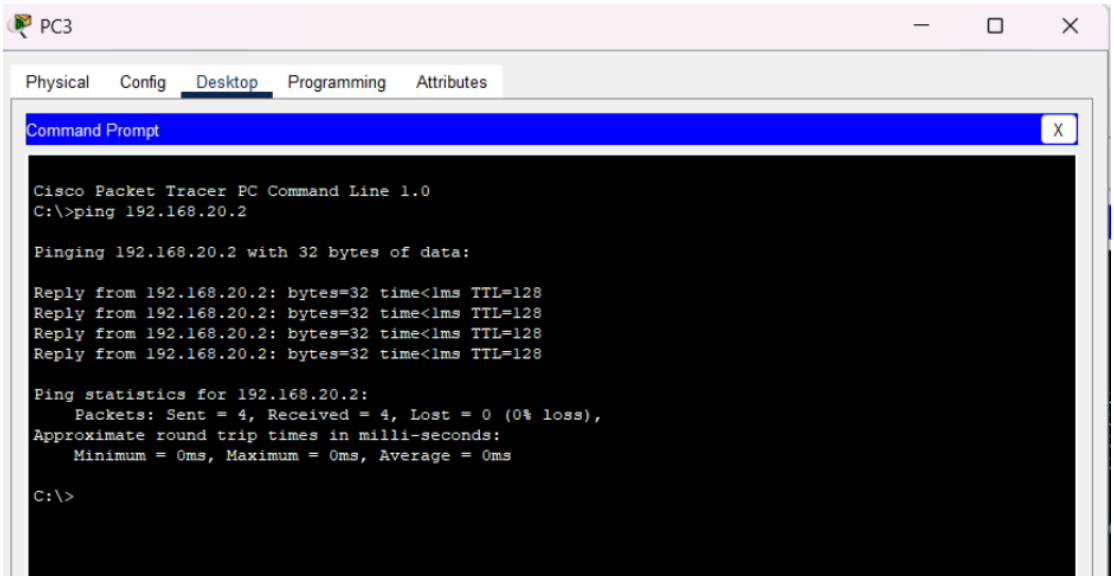


Ping from all PCs to Web server

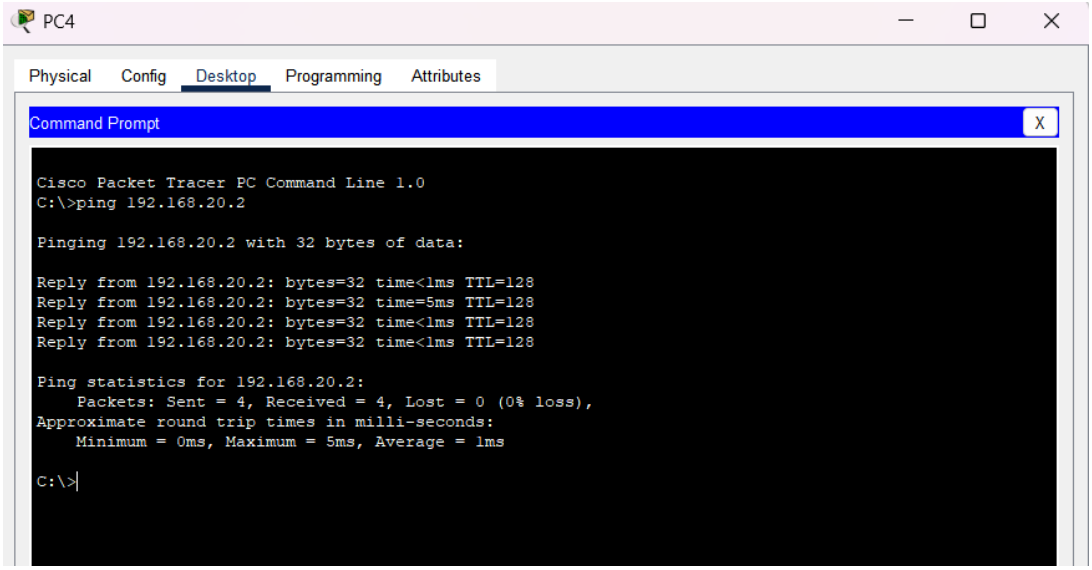
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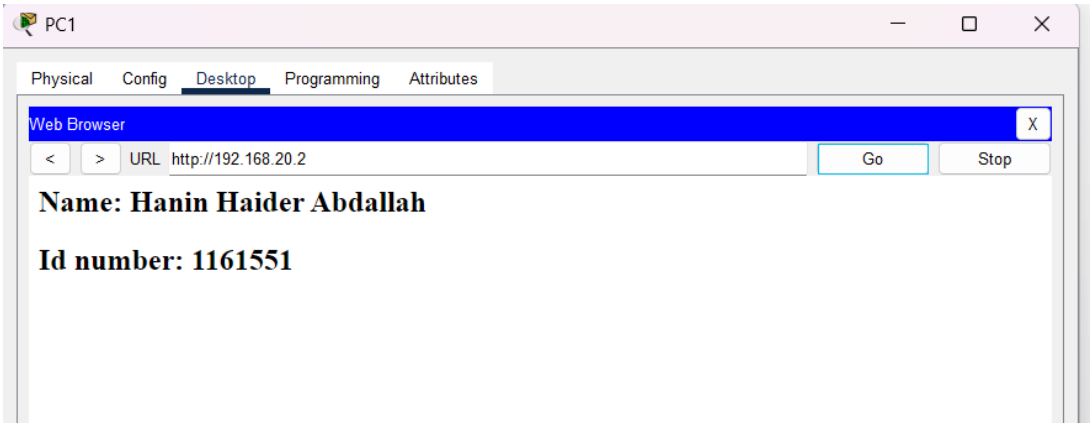




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Access to the Web Servers homepage from all PCs



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Critical reflection

My network follows a Star topology which can be expensive when operating for a bigger business or program network, because a more professional network requires more than just 2 servers a few PCs to run which takes longer time to build and hence more money to build but also to maintain. Another issue is the lack of reliability due to only having one router and one switch meaning that if one device fails in my network then my entire network would collapse and not function, therefore a solution would be to implement more routers and switches which is much more suitable for a real life network environment. Another limitation is that the DHCP server is only available in the VLAN 10 while in a real world example there would be more than just one DHCP server that is

connected to different department or networks rather than to just one VLAN 10 as it would make it more practical to operate automatically address all VLANs or configure multiple DHCP servers and risk free. Lastly, one more limitation to my network is that I have not included an DNS server which means that it is not suitable or practicable for a network for a real-world organisation where the user would need to remember every IP address to access the Web which limits the efficiency **[Chiradeep BasuMallick, 2022]**

In a real-life LAN for a campus or a company I would instead:
Configure a multi cell Wlan to allow wireless connections as in real world only relying on wired network or in campus is not practical or realistic.
I would also add more VLANs for each department for better security as the information would only send to the right department e.g., The finance department would not have access to the same data as the HR department. and, to reduce broadcast traffic.
[Amin Amini, Lecture 9]

Use a layer 3 switch as the latency is lower, as the data does not need to leave the switch to be routed to a different network and they are commonly used in lans which makes them perfect for a real life VLAN such on a campus When it comes to real life organisations may have classified or private data so it's important to implement security masseurs such as security system to prevent unauthorised access into or out of the network. But also use HTTPS to encrypt the data during communication.
[Cisco Networking Academy, 2020]

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